Two-step PCR amplicon library workflow

1st PCR adds head seq → 2nd PCR adds barcodes → pool → Nanopore (Evoprep) · HuangLab LIMS stages 1 → 3a → 3b → 4a

Stage 1 — Specimen# / Culture#

Stage 3a — extraction + Qubit · MolWork#

Stage 3b — PCR + gel + Qubit · pcr_reaction row

Stage 4a — Nanopore experiment · Nanopore#

Specimen registered in LIMS
soil / environmental → Specimen# | fungal strain → Culture#

LIMS register
`/lims-register`
Stage 1 → Specimen# (soil / env.) or Culture# (strain)

Soil DNA extraction
SuperSoil protocol

Fungal DNA extraction
modified BOMB v9.0

DNA extraction

DNA template → MolWork#
soil crude extract (SuperSoil) | pure fungal extraction (BOMB v9)
MolWork# minted after quantification

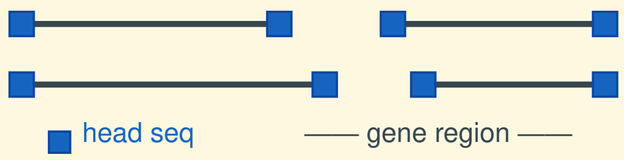


Qubit Flex
quantify the extract (Stage 3a)

LIMS register
`/lims-register`
Stage 3a → MolWork# minted after quantification

1st PCR (gene regions)

1st PCR products (per sample)



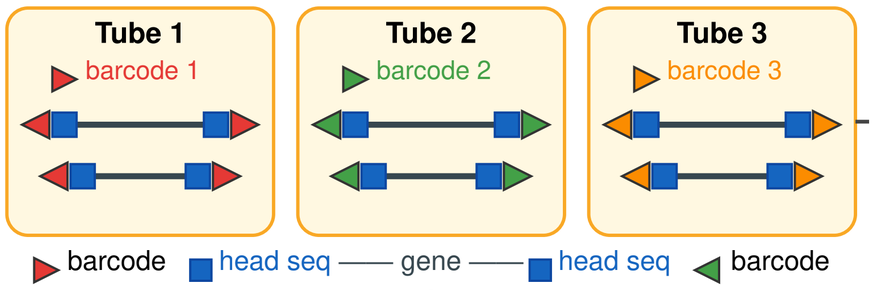


Gel check #1
keep only good bands

2-step PCR protocol
mixtures + cycling conditions

2nd PCR (barcoding)
good amplicons only

2nd PCR products: one tube per sample, one barcode per tube





Gel check #2
confirm barcoded product



Qubit Flex
quantify → 10 ng/µL (Stage 3b)

Pooling (equal molar)

Pool all barcoded samples
equal molar of every sample

conc. → equal-molar volumes

LIMS register
`/lims-register`
Stage 3b → pcr_reaction: conc., 2nd_pcr_ok, barcode



Pooled sample

Size selection 1X
bead purification protocol

Purification (bead clean-up)
PCR sample : bead = 1 : 1



Purified pooled sample

EvoPrep library prep
KAPA EvoPrep NBD114.96

Ready to be sequenced
library construction using Evoprep

LIMS register
`/lims-register`
Stage 4a → Nanopore#: experiment, samples, NanoAmp

Legend

- Head seq (universal tail on the gene-region primers)
- Barcode (sample-specific, on the barcode primers)
- Gene region (target amplicon; colour = different sample)

QR codes (left) open the lab's self-hosted protocol PDF for that step
left bands = LIMS stages; /lims-register at every stage; dashed = QC / info flow